MATH120: Functions and Graphs

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Learning Objectives

- Evaluate a function using function notation f(x)
- Determine the domain and range of a function
- Compute slope and write the equation of a line
- Find the composition and inverse of a function

Simplify each expression completely. Show all steps and circle your final answer.

Domain of functions

1. Find the domain of g(x) = 1 / (x - -5). State any values of x that must be excluded.

$$g(x) = \frac{1}{x - -5}$$

Answer: ____

2. Find the domain of g(x) = 1 / (x - 1). State any values of x that must be excluded.

$$g(x) = \frac{1}{x - 1}$$

Answer: ____

3. Find the domain of g(x) = 4 / (x - -1). State any values of x that must be excluded.

$$g(x) = \frac{4}{x - -1}$$

Answer: ____

4. Find the domain of g(x) = 2 / (x - 2). State any values of x that must be excluded.

$$g(x) = \frac{2}{x - 2}$$

Answer: ____

5. Find the domain of g(x) = 5 / (x - -2). State any values of x that must be excluded.

$$g(x) = \frac{5}{x - -2}$$

Answer: ____

6. Find the domain of g(x) = 5 / (x - -3). State any values of x that must be excluded.

$$g(x) = \frac{5}{x - -3}$$

Answer: ____

7. Find the domain of g(x) = 1 / (x - -1). State any values of x that must be excluded.

$$g(x) = \frac{1}{x - -1}$$

Answer: ____

8. Find the domain of g(x) = 1 / (x - -1). State any values of x that must be excluded.

$$g(x) = \frac{1}{x - -1}$$

Answer: ____

Evaluating functions

9. A company's revenue is modeled by R(x) = 12x + 246, where x is units sold. Find R(9).

$$f(x) = 12x + 246, \quad x = 9$$

Answer: ____

10. A profit function is P(x) = -1x^2 + 183. Find P(6).

$$-1x^2 + 183,\ x = 6$$

Answer: ____

11. A cost function is C(x) = 13x + 627 dollars, where x is the number of items produced. Find C(25) and interpret the result.

$$f(x) = 13x + 627, \quad x = 25$$

Answer: ____

12. A company's revenue is modeled by R(x) = 32x + 132, where x is units sold. Find R(18).

$$f(x) = 32x + 132, \quad x = 18$$

Answer: ____

13. A profit function is P(x) = -2x^2 + 83. Find P(6).

$$-2x^2 + 83,\ x = 6$$

Answer: ____

14. A cost function is C(x) = 7x + 552 dollars, where x is the number of items produced. Find C(23) and interpret the result.

$$f(x) = 7x + 552, \quad x = 23$$

Answer: ______________________________________

15. A company's revenue is modeled by R(x) = 25x + 357, where x is units sold. Find R(10).

$$f(x) = 25x + 357, \quad x = 10$$

Answer: ______________________________________

16. A profit function is P(x) = -2x^2 + 63. Find P(6).

$$-2x^2 + 63, \ x = 6$$

Answer: ______________________________________

17. A cost function is C(x) = 24x + 421 dollars, where x is the number of items produced. Find C(24) and interpret the result.

$$f(x) = 24x + 421, \quad x = 24$$

Answer: ______________________________________

18. A company's revenue is modeled by R(x) = 15x + 200, where x is units sold. Find R(9).

$$f(x) = 15x + 200, \quad x = 9$$

Answer: ______________________________________

19. A profit function is P(x) = -4x^2 + 171. Find P(6).

$$-4x^2 + 171, \ x = 6$$

Answer: ______________________________________

20. A cost function is C(x) = 6x + 576 dollars, where x is the number of items produced. Find C(28) and interpret the result.

$$f(x) = 6x + 576, \quad x = 28$$

Answer: ______________________________________

21. A company's revenue is modeled by R(x) = 24x + 143, where x is units sold. Find R(19).

$$f(x) = 24x + 143, \quad x = 19$$

Answer: ______________________________________

22. A profit function is P(x) = -2x^2 + 122. Find P(6).

$$-2x^2 + 122,\ x = 6$$

Answer: ______________________________

23. A cost function is C(x) = 20x + 386 dollars, where x is the number of items produced. Find C(27) and interpret the result.

$$f(x) = 20x + 386, \quad x = 27$$

Answer: ______________________________

24. A company's revenue is modeled by R(x) = 38x + 327, where x is units sold. Find R(14).

$$f(x) = 38x + 327, \quad x = 14$$

Answer: ______________________________

25. A profit function is P(x) = -1x^2 + 81. Find P(6).

$$-1x^2 + 81,\ x = 6$$

Answer: ______________________________

26. A cost function is C(x) = 18x + 495 dollars, where x is the number of items produced. Find C(19) and interpret the result.

$$f(x) = 18x + 495, \quad x = 19$$

Answer: ______________________________

27. A company's revenue is modeled by R(x) = 14x + 159, where x is units sold. Find R(18).

$$f(x) = 14x + 159, \quad x = 18$$

Answer: ______________________________

28. A profit function is P(x) = -1x^2 + 178. Find P(2).

$$-1x^2 + 178,\ x = 2$$

Answer: ______________________________

29. A cost function is C(x) = 26x + 238 dollars, where x is the number of items produced. Find C(13) and interpret the result.

$$f(x) = 26x + 238, \quad x = 13$$

Answer: ______________________________

30. A company's revenue is modeled by R(x) = 30x + 360, where x is units sold. Find R(5).

$$f(x) = 30x + 360, \quad x = 5$$

Answer: ____

Topics: Evaluating functions, Domain of functions. All answers verified by independent computation.

Solutions

Domain of functions

1. Find the domain of g(x) = 1 / (x - -5). State any values of x that must be excluded.

$$g(x) = \frac{1}{x - -5}$$

→ Set denominator not equal to zero: x - -5 ≠ 0 → x ≠ -5.
→ Domain: all real numbers except x = -5.

Answer: $x \neq -5$

2. Find the domain of g(x) = 1 / (x - 1). State any values of x that must be excluded.

$$g(x) = \frac{1}{x - 1}$$

→ Set denominator not equal to zero: x - 1 ≠ 0 → x ≠ 1.
→ Domain: all real numbers except x = 1.

Answer: $x \neq 1$

3. Find the domain of g(x) = 4 / (x - -1). State any values of x that must be excluded.

$$g(x) = \frac{4}{x - -1}$$

→ Set denominator not equal to zero: x - -1 ≠ 0 → x ≠ -1.
→ Domain: all real numbers except x = -1.

Answer: $x \neq -1$

4. Find the domain of g(x) = 2 / (x - 2). State any values of x that must be excluded.

$$g(x) = \frac{2}{x - 2}$$

→ Set denominator not equal to zero: x - 2 ≠ 0 → x ≠ 2.
→ Domain: all real numbers except x = 2.

Answer: $x \neq 2$

5. Find the domain of g(x) = 5 / (x - -2). State any values of x that must be excluded.

$$g(x) = \frac{5}{x - -2}$$

→ Set denominator not equal to zero: x - -2 ≠ 0 → x ≠ -2.
→ Domain: all real numbers except x = -2.

Answer: $x \neq -2$

6. Find the domain of g(x) = 5 / (x - -3). State any values of x that must be excluded.

$$g(x) = \frac{5}{x - -3}$$

→ Set denominator not equal to zero: x - -3 ≠ 0 → x ≠ -3.
→ Domain: all real numbers except x = -3.

Answer: $x \neq -3$

7. Find the domain of g(x) = 1 / (x - -1). State any values of x that must be excluded.

$$g(x) = \frac{1}{x - -1}$$

→ *Set denominator not equal to zero: x - -1 ≠ 0 → x ≠ -1.*
→ *Domain: all real numbers except x = -1.*

Answer: $x \neq -1$

8. Find the domain of g(x) = 1 / (x - -1). State any values of x that must be excluded.

$$g(x) = \frac{1}{x - -1}$$

→ *Set denominator not equal to zero: x - -1 ≠ 0 → x ≠ -1.*
→ *Domain: all real numbers except x = -1.*

Answer: $x \neq -1$

Evaluating functions

9. A company's revenue is modeled by R(x) = 12x + 246, where x is units sold. Find R(9).

$$f(x) = 12x + 246, \quad x = 9$$

→ *Substitute x = 9: R(9) = 12(9) + 246.*
→ *= 108 + 246 = 354.*

Answer: $f(9) = 354$

10. A profit function is P(x) = -1x^2 + 183. Find P(6).

$$-1x^2 + 183, \ x = 6$$

→ *Substitute x = 6: P(6) = -1(6)^2 + 183.*
→ *= -1(36) + 183 = -36 + 183 = 147.*

Answer: $-1(6)^2 + 183 = -36 + 183 = 147$

11. A cost function is C(x) = 13x + 627 dollars, where x is the number of items produced. Find C(25) and interpret the result.

$$f(x) = 13x + 627, \quad x = 25$$

→ *C(25) = 13(25) + 627 = 325 + 627 = 952.*
→ *It costs $952 to produce 25 items.*

Answer: $f(25) = 952$

12. A company's revenue is modeled by R(x) = 32x + 132, where x is units sold. Find R(18).

$$f(x) = 32x + 132, \quad x = 18$$

→ *Substitute x = 18: R(18) = 32(18) + 132.*
→ *= 576 + 132 = 708.*

Answer: $f(18) = 708$

13. A profit function is P(x) = -2x^2 + 83. Find P(6).

$$-2x^2 + 83, \ x = 6$$

→ *Substitute x = 6: P(6) = -2(6)^2 + 83.*
→ *= -2(36) + 83 = -72 + 83 = 11.*

Answer: $-2(6)^2 + 83 = -72 + 83 = 11$

14. A cost function is C(x) = 7x + 552 dollars, where x is the number of items produced. Find C(23) and interpret the result.

$$f(x) = 7x + 552, \quad x = 23$$

→ *C(23) = 7(23) + 552 = 161 + 552 = 713.*
→ *It costs $713 to produce 23 items.*

Answer: $f(23) = 713$

15. A company's revenue is modeled by R(x) = 25x + 357, where x is units sold. Find R(10).

$$f(x) = 25x + 357, \quad x = 10$$

→ *Substitute x = 10: R(10) = 25(10) + 357.*
→ *= 250 + 357 = 607.*

Answer: $f(10) = 607$

16. A profit function is P(x) = -2x^2 + 63. Find P(6).

$$-2x^2 + 63,\ x = 6$$

→ *Substitute x = 6: P(6) = -2(6)^2 + 63.*
→ *= -2(36) + 63 = -72 + 63 = -9.*

Answer: $-2(6)^2 + 63 = -72 + 63 = -9$

17. A cost function is C(x) = 24x + 421 dollars, where x is the number of items produced. Find C(24) and interpret the result.

$$f(x) = 24x + 421, \quad x = 24$$

→ *C(24) = 24(24) + 421 = 576 + 421 = 997.*
→ *It costs $997 to produce 24 items.*

Answer: $f(24) = 997$

18. A company's revenue is modeled by R(x) = 15x + 200, where x is units sold. Find R(9).

$$f(x) = 15x + 200, \quad x = 9$$

→ *Substitute x = 9: R(9) = 15(9) + 200.*
→ *= 135 + 200 = 335.*

Answer: $f(9) = 335$

19. A profit function is P(x) = -4x^2 + 171. Find P(6).

$$-4x^2 + 171,\ x = 6$$

→ *Substitute x = 6: P(6) = -4(6)^2 + 171.*
→ *= -4(36) + 171 = -144 + 171 = 27.*

Answer: $-4(6)^2 + 171 = -144 + 171 = 27$

20. A cost function is C(x) = 6x + 576 dollars, where x is the number of items produced. Find C(28) and interpret the result.

$$f(x) = 6x + 576, \quad x = 28$$

→ *C(28) = 6(28) + 576 = 168 + 576 = 744.*
→ *It costs $744 to produce 28 items.*

Answer: $f(28) = 744$

21. A company's revenue is modeled by R(x) = 24x + 143, where x is units sold. Find R(19).

$$f(x) = 24x + 143, \quad x = 19$$

→ *Substitute x = 19: R(19) = 24(19) + 143.*

→ *= 456 + 143 = 599.*

Answer: $f(19) = 599$

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22. A profit function is P(x) = -2x^2 + 122. Find P(6).

$$-2x^2 + 122, \ x = 6$$

→ *Substitute x = 6: P(6) = -2(6)^2 + 122.*

→ *= -2(36) + 122 = -72 + 122 = 50.*

Answer: $-2(6)^2 + 122 = -72 + 122 = 50$

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23. A cost function is C(x) = 20x + 386 dollars, where x is the number of items produced. Find C(27) and interpret the result.

$$f(x) = 20x + 386, \quad x = 27$$

→ *C(27) = 20(27) + 386 = 540 + 386 = 926.*

→ *It costs $926 to produce 27 items.*

Answer: $f(27) = 926$

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24. A company's revenue is modeled by R(x) = 38x + 327, where x is units sold. Find R(14).

$$f(x) = 38x + 327, \quad x = 14$$

→ *Substitute x = 14: R(14) = 38(14) + 327.*

→ *= 532 + 327 = 859.*

Answer: $f(14) = 859$

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25. A profit function is P(x) = -1x^2 + 81. Find P(6).

$$-1x^2 + 81, \ x = 6$$

→ *Substitute x = 6: P(6) = -1(6)^2 + 81.*

→ *= -1(36) + 81 = -36 + 81 = 45.*

Answer: $-1(6)^2 + 81 = -36 + 81 = 45$

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26. A cost function is C(x) = 18x + 495 dollars, where x is the number of items produced. Find C(19) and interpret the result.

$$f(x) = 18x + 495, \quad x = 19$$

→ *C(19) = 18(19) + 495 = 342 + 495 = 837.*

→ *It costs $837 to produce 19 items.*

Answer: $f(19) = 837$

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27. A company's revenue is modeled by R(x) = 14x + 159, where x is units sold. Find R(18).

$$f(x) = 14x + 159, \quad x = 18$$

→ *Substitute x = 18: R(18) = 14(18) + 159.*
→ *= 252 + 159 = 411.*

Answer: $f(18) = 411$

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28. A profit function is P(x) = -1x^2 + 178. Find P(2).

$$-1x^2 + 178, \ x = 2$$

→ *Substitute x = 2: P(2) = -1(2)^2 + 178.*
→ *= -1(4) + 178 = -4 + 178 = 174.*

Answer: $-1(2)^2 + 178 = -4 + 178 = 174$

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29. A cost function is C(x) = 26x + 238 dollars, where x is the number of items produced. Find C(13) and interpret the result.

$$f(x) = 26x + 238, \quad x = 13$$

→ *C(13) = 26(13) + 238 = 338 + 238 = 576.*
→ *It costs $576 to produce 13 items.*

Answer: $f(13) = 576$

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30. A company's revenue is modeled by R(x) = 30x + 360, where x is units sold. Find R(5).

$$f(x) = 30x + 360, \quad x = 5$$

→ *Substitute x = 5: R(5) = 30(5) + 360.*
→ *= 150 + 360 = 510.*

Answer: $f(5) = 510$
